Accepted Manuscript

Submission Date: 2023-06-08
Accepted Date: 2024-03-05
Accepted Manuscript online: 2024-03-05

International Journal of Sports Medicine

The acute:chronic workload ratio and injury risk in semiprofessional football players

Roberto Seco-Serna, Carlos Lago-Fuentes, Martín Barcala-Furelos.

Affiliations below.

DOI: 10.1055/a-2282-0024

Please cite this article as: Seco-Serna R, Lago-Fuentes C, Barcala-Furelos M. The acute:chronic workload ratio and injury risk in semi-professional football players. International Journal of Sports Medicine 2024. doi: 10.1055/a-2282-0024

Conflict of Interest: The authors declare that they have no conflict of interest.

Abstract:
The purpose of this study was to analyse the association and predictive capacity between the acute:chronic workload ratio (ACWR) and non-contact injuries in a semiprofessional football team. 17 football players from a Spanish Third Division football team participated voluntarily in this study. A prospective longitudinal study was developed during 2020/2021 season. 24 weeks were analysed from October to March, including a regenerative microcycle due to the absence of competition during Christmas. Ratio perceived exertion (RPE) and session- ratio perceived exertion (sRPE) were registered for every training and game session. After that, acute and chronic workloads were calculated, as well as ACWR was derived from them. Furthermore, non-contact injuries were registered during the mentioned period. The main findings were: there is no significant association between the ACWR and non-contact injuries (0.069 ($p<0.05$)) and, the use of the ACWR by itself is insufficient to predict the occurrence of non-contact injuries in a semiprofessional football team. Consequently, the ACWR is not an useful predictive tool for injuries in semiprofessional football teams.

Corresponding Author:
Dr. Carlos Lago-Fuentes, Universidad Europea del Atlantico Facultad de Ciencias de la Salud, Faculty of Education and Sports Sciences, Calle Isabel Torres,21, 39011 Santander, Spain, carloslagofuentes@hotmail.com

Affiliations:
Roberto Seco-Serna, Universidad Europea del Atlántico Facultad de Ciencias de la Salud, Sports Sciences, Santander, Spain
Carlos Lago-Fuentes, Universidad Europea del Atlantico Facultad de Ciencias de la Salud, Faculty of Education and Sports Sciences, Santander, Spain
Martín Barcala-Furelos, Universidad Europea del Atlantico Facultad de Ciencias de la Salud, Sport Sciences, Santander, Spain

This article is protected by copyright. All rights reserved.

Downloaded by: University of Cincinnati. Copyrighted material.

Accepted Manuscript

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Thieme

The acute:chronic workload ratio and injury risk in semiprofessional football players.

Abstract

The purpose of this study was to analyse the association and predictive capacity between the acute:chronic workload ratio (ACWR) and non-contact injuries in a semiprofessional football team. 17 football players from a Spanish Third Division football team participated voluntarily in this study. A prospective longitudinal study was developed during 2020/2021 season. 24 weeks were analysed from October to March, including a regenerative microcycle due to the absence of competition during Christmas. Rate perceived exertion (RPE) and session- rate perceived exertion (sRPE) were registered for every training and game session. After that, acute and chronic workloads were calculated, and ACWR was subsequently derived from them. Furthermore, non-contact injuries were registered during the mentioned period. The main findings were: there is a poor correlation between the ACWR and non-contact injuries ($r=0.069$ ($p<0.05$)) and, the use of the ACWR by itself is insufficient to predict the occurrence of non-contact injuries in a semiprofessional football team. Consequently, the ACWR is not an useful predictive tool for injuries in semiprofessional football teams.

Keywords: load, non-contact injuries, team sport, predictive tools

Introduction

Currently, one of the main objectives of physical preparation in any type of sport is to promote the full availability and well-being of players to the greatest extent possible and thus

This article is protected by copyright. All rights reserved.

Downloaded by: University of Cincinnati. Copyrighted material.

Accepted Manuscript

reduce the risk of injury [1]. However, despite the efforts carried out to meet this objective in both human and material terms, a high number of injuries are recorded each year in team sports like football [2], which is a major problem for clubs in both sporting and economic terms. In football nowadays, although the injury rate is going down [3], the total number of specific types of injuries, such as hamstring injuries, has actually increased in recent years [4]. In any case, the reality is that having fewer injuries during the season is related to greater sporting success [5, 6], so clubs should strive to meet this objective.

Faced with this issue, science provides a range of tools with a solid scientific backing which have proved to be effective in minimizing both the appearance and severity of injuries. Different tools can be applied in different football contexts in order to reduce the number and severity of injuries. Some of the means that can be relied on to achieve these objectives include the following: workload control using GPS [7, 8], the use of assessment tests [9], warm-up/activation protocols [10] or other recovery strategies [11].

Quantification of training load and injury prevention require both human and material resources and sometimes such costs are unaffordable for many clubs, due to economic limitations. In this sense, there are alternatives that can help in the search for solutions. The Subjective Perception of Effort [SPE] [12] is a tool for quantifying the training workload without any economic cost and is applicable in non-professional contexts which can facilitate the quantification of the training workload without the need for significant resources [13]. Despite being a subjective tool, it has been validated in recent decades due to its high correlation with different internal workload variables, such as heart rate or oxygen consumption [13, 14]. In addition, it is also valid for external workload measures such as total distance travelled or player workload [14, 15].

In this way, the acute:chronic workload ratio [ACWR] is purported to predict injury risk. It yields a training workload value and the player increases or decreases his/her chances

This article is protected by copyright. All rights reserved.
Downloaded by: University of Cincinnati. Copyrighted material.
Accepted Manuscript

of getting injured depending on that value [16]. This model considers the risk that an athlete has of getting injured, based on the relationship between their current state of fitness (chronic workload) and their level of fatigue (acute workload) [16]. According to Gabbett [16], as long as said ratio remains between a value of 0.8 and 1.3, the player is considered to be within the "sweet spot", i.e. having an optimal ACWR to minimize the potential risk of injury. On the other hand, when the ratio exceeds 1.5 or falls below 0.8, the player enters the "red zone" and the risk of suffering an injury increases.

The predictive capacity of the ACWR has been studied by sports scientists since its inception and several studies support the idea that this ratio is a good predictor of the risk of being injured. The reason is because their results show that, for example, ACWR values >2.00 increase the risk of getting injuries [17], ratio values between 1.00 and 1.25 minimize the risk of injury [18], peaks in training workload increase the likelihood of injuries [19] and workload peaks in the process of returning to competition increase the risk of recurrence [20]. However, and in contrast to these publications, other studies have reached different conclusions and there is currently a strong undercurrent which strongly opposes the idea that the ratio has a good predictive capacity to detect the occurrence of future injuries [21]. These state the poor ability of the ACWR to predict the occurrence of future injuries [22, 23] and discourage its use in the planning of teams and amongst athletes [24]. To date, several researchers [16, 18, 22] have developed studies on this topic in professional football teams. However, previous studies have not analyzed the use of this tool in semiprofessional teams (TIER 3) and, because it is a very low-cost tool, the study and application in semiprofessional contexts is of great interest. Because of all these reasons, the objective of this study was to analyse the association and predictive capacity between the ACWR and non-contact injuries in a semiprofessional football team.

This article is protected by copyright. All rights reserved.
Downloaded by: University of Cincinnati. Copyrighted material.
Accepted Manuscript

Methods

Study design

The study data was collected during the competitive period of the 2020/2021 season, and excluded the pre-season, second league and play-off matches. In total, 24 weeks were analysed from October to March, including a regenerative microcycle due to the absence of competition during Christmas. The team in which the data collection was carried out did their usual training of 3 days a week on the pitch and none in the gym from Monday to Friday and played a game at the weekend.

The procedures used in this study included an informed consent form, which the players had to fill out at the beginning of the study, and an intervention evaluation request. Both documents were approved by the research ethics committee of the university. (CEI-8655/2021).

Participants

A prospective observational study was carried out during the 2020/2021 season, in which the sample was non-probabilistic and consisted of 17 football players from a Spanish Third Division football team (4th national category) from group III (the Cantabria region), which corresponded to “TIER” 3 [25]. All the participants were of Spanish nationality. Table 1 shows the descriptive analysis of the participants. Goalkeepers were excluded and only outfield players were considered.

*****Insert Table 1 near here*****

Procedures

At the end of each training or league game, the players gave the value of The Rate of Perceived Exertion (RPE), according to the modified Borg scale [26]. The length of time each

This article is protected by copyright. All rights reserved.
Accepted Manuscript
Downloaded by: University of Cincinnati. Copyrighted material.

football player participated in it was also recorded by the investigator. The RPE was collected anonymously and personally for each player, using the MindPro [Valladolid, Spain] mobile app, which allows, and this was done, this value to be collected individually on a daily basis for the other teammates, and is manifested by the players between 15 minutes and up to 48 hours after the end of the training session or the match, because even if it collected 48 hours after the end of the session, it is still valid [27]. As for the duration of the sessions, they were also recorded daily. In the event that a player did not complete the session, the amount of time that player was training or competing was recorded. Subsequently, the data, time and RPE were all transferred to a digital registry.

After recording both types of data, the sRPE was calculated individually [26]. Likewise, the acute workload (N=7); where *N* means "number of days", and chronic workload (N=28) was calculated for the entire competitive period recorded, using the mathematical formula presented below to finally obtain the ACWR by dividing the acute load by the chronic load.

The ACWR was obtained using the "exponentially weighted moving averages" (EWMA) model described by Williams et al. [28] and which has been shown to be more sensitive than the traditional model when it comes to predicting the risk of getting injuries [29]. This model uses the following mathematical formula:

$$EWMA_{today} = Load_{today}\ x\ \lambda_a + ((1 - \lambda_a)\ x\ EWMA_{yesterday}$$

Where: λ_a=2/(N+1), with N being the number of days used to calculate the acute and chronic workload. In the present study, N was equal to 7 for the calculation of the acute load and N was equal to 28 for the calculation of the chronic workload.

Only injuries which incurred a loss of time [30] and which were caused without contact during training sessions or matches were considered, omitting all injuries occurring at home, the workplace or at any other time or place that was not during the football practice.

This article is protected by copyright. All rights reserved.
Accepted Manuscript
Downloaded by: University of Cincinnati. Copyrighted material.

Injuries were recorded anecdotally at the time they occurred, using a log in which the researcher recorded the variables to be considered (contact/non-contact, practicing football/other, minute of game/training in which it occurred), through direct observation and complementing it, if it was necessary, by asking the staff and/or players.

Return to the group was defined as the first day after the injury in which the player completed a full training session with the group or participated in an official match [31]. The days that a player was injured were calculated by counting the days elapsed from the day the injury occurred until the day they returned to the group.

Statistical analysis

Players were only included in the statistical analysis when they were healthy, that is, free from injury. Whenever a player picked up an injury, he was excluded from the day after he was injured until he returned to complete a training session with the rest of the team. Thus, by participating fully and completely that player then became part of the analysis once again. IMB SPSS version 25.0 software was used for the statistical analysis by means of performing a Spearman correlation test. In addition, the total number of true positives, false positives, true negatives, and false negatives, as well as the sensitivity and specificity of the test were calculated. These parameters were defined in accordance with Kent et al. [32].

The following is how the cases were considered: "true positive" cases were when the player had an injury with a ACWR ≥1.5 or <0.8; “false positive” cases were those with a ACWR ≥1.5 or <0.8, i.e. the player was not injured; “true negative” cases were defined as having a ACWR <1.50 or ≥0.8, i.e. the player was not injured; “false negative” cases were those with a ACWR <1.50 or ≥0.8, i.e. the player was injured. Sensitivity and specificity were calculated as follows [22]:

$$Sensitivity = \frac{True\ positives}{(True\ positives + False\ negatives)} * 100$$

This article is protected by copyright. All rights reserved.
Accepted Manuscript
Downloaded by: University of Cincinnati. Copyrighted material.

$$Specificity = \frac{False\ positives}{(False\ positives + True\ negatives)} * 100$$

Results

A total of 10 non-contact injuries were sustained by the players during the weeks of the season included in the study and most of them were muscular injuries. The injury incidence of non-contact injuries during the study period was 5.91 per 1000 hours of practice (i.e. training and matches), while the severity was 11.90 ± 10.38 days out injured. Table 2 presents a summary of the number and type of injuries that occurred, as well as their incidence and severity.

*****Insert Table 2 near here*****

Table 3 shows the results obtained after performing a Spearman correlation test. A very low correlation of 0.069 (p = 0.000) was obtained between the value of the ACWR and the occurrence of a non-contact injury, that is, the relationship between both variables is almost non-existing. In addition, Figure 1 allows us to observe how the distribution of injuries was based on the value of the ACWR throughout the 24 weeks included.

*****Insert Table 3 near here*****

*****Insert Figure 1 near here*****

Finally, Table 4 shows the frequency analysis according to the 4 cases described in the statistical analysis of the number of cases in which the Distribution of non-contact injuries and contact injuries according to the value of the ACWR was able to correctly detect the occurrence of an injury or not, based on whether this was ≥1.5 or <0.8. According to this, the

This article is protected by copyright. All rights reserved.

Downloaded by: University of Cincinnati. Copyrighted material.

Accepted Manuscript

ACWR alongside the 24 weeks of the study showed only 5 true positives were reported, that is, only 5 injuries were reported with a ACWR lower than 0.8 or upper than 1.5.

*****Insert Table 4 near here*****

Discussion

To date, this is the first ever study to verify the association and predictive capacity of the ACWR with the risk of injury in semiprofessional football players. The objective of this study was to analyse the association and predictive capacity between the ACWR workload ratio and non-contact injuries in a semiprofessional football team. In relation to this, the main findings were: a) there is no significant association between the ACWR and non-contact injuries and b) the use of the ACWR by itself is insufficient to predict the occurrence of non-contact injuries in a semiprofessional football team.

The team studied had an injury incidence of 8.86 injuries per 1000 hours of practice during the analysed season (Table 2). This value is very close to the one presented by Beijsterveldt et al. [33], who obtained an injury incidence of 9.6 in amateur football players. However, there are other non-professional environments studied which have reported significantly lower incidences. For example, Hägglund et al. [34] found an incidence among their football players of 5.2 injuries per thousand hours of practice. Regarding the types of injuries suffered, Beijsterveldt et al. [33] and Hägglund et al. [34] reported 23.6% and 14.3% of ligamentous injuries respectively, which are values similar to the 20% of this particular type of injury obtained in the present study. In addition, muscular injuries at 70%, and tendon injuries at 10% vary meaningfully from the percentages found by these authors for these types of injuries: 48.4% muscular, 43.8% tendon [34] and 38.9% muscle-tendon [33]. These differences are probably due to the fact that the time period studied in these works was longer than the period analysed in the present research, which suggests that if a longer follow-up

This article is protected by copyright. All rights reserved.
Accepted Manuscript
Downloaded by: University of Cincinnati. Copyrighted material.

were carried out, the percentage of each type of injury obtained would possibly be similar to a large degree.

In order to respond to the main objective of the study, the sensitivity of the test was obtained, which is defined as the ability of a test to detect a positive when the subject is positive [i.e. true positive] and the specificity of the test, which is defined as the ability of a test to detect a negative when the subject is negative (i.e. true negative) [32]. The sensitivity and specificity (see Table 4) obtained were 50.00% and 31.35% respectively, which are results that show low predictive capacity when detecting non-contact injuries using the ACWR as a predictor. This result supports the conclusion of Fanchini et al. [22], who state that this tool has a poor predictive ability when it comes to detecting future non-contact injuries in football players. They found a sensitivity of 20.0% which opposes the one shown by Gabbett [16], who obtained a sensitivity of 87.1% and a specificity of 98.8%.

On the other hand, in relation to the association between the ACWR and the risk of non-contact injury, Fanchini et al. [22] found that the ACWR is positively related to a higher risk of injury, that is, the higher the ACWR, the greater the risk of getting a non-contact injury. The correlation test applied in the present study (see table 3) shows that there is no relationship between the ACWR and the occurrence of injuries. However, once again the shorter time period studied may be one of the reasons why such a relationship does not appear, since the total number of non-contact injuries suffered is lower [n=10] than, for example, in the study by Fanchini et al. (n=90) [22], or in Gabbet´s study (n=139) [16].

In any case, the results show that the predictive capacity of the ACWR is weak, which is in line with one section of the scientific community [21–24], who currently strongly rejects this idea, arguing that the values of the ratio are based on erroneous conceptual premises [24], both from a mathematical and practical point of view, concluding that the recommendations on its use within team planning must be updated and corrected in order to avoid its use [21].

This article is protected by copyright. All rights reserved.

Downloaded by: University of Cincinnati. Copyrighted material.

Accepted Manuscript

Despite the fact that the present investigation has found a weak relationship between the ACWR and the occurrence of injuries (table 3), in addition to poor predictive capacity to detect future injuries (table 4), other studies have been able to associate both variables. Said studies find, for instance, that an ACWR between 1.00 and 1.25 reduces the risk of suffering a non-contact injury [18], that ACWR ≥1.5 increase the probability of non-contact injury [16, 18] and that large spikes in acute workload or sudden increases in workload correspond to an increased risk of non-contact injury [19]. In spite of these findings, and as pointed out by Griffin et al. [23], the fact that there is an association between the ACWR and non-contact injuries does not mean that these can be predicted using this method, since the relationship and the predictive ability are two different things.

However, although no association has been found between the ACWR and the risk of non-contact injury, in recent years recent studies have shown how the weekly organization of the load influences the risk of injury, especially in periods of high competitive congestion [35–37], so targeting the use of the ACWR to this goal may be useful since, when football players suffer non-contact injuries, such as anterior cruciate ligament (ACL) or common muscle injuries in football like hamstring muscle injuries, several risk factors environmental, anatomical, hormonal, neuromuscular, biomechanical, hours of exposure (both train and match) are known that predispose the athlete to having a greater or lesser risk of getting them [38–41]. Technical staff works on such areas so as to try to eliminate them or, at least, reduce their presence to the greatest extent possible in each of the players. Some of these risk factors are related to the training workload, which can lead to a greater or lesser risk of getting an injury depending on how it is administered [42]. It is at this point when the ACWR can be used as a tool in injury prevention, since it enables regulation of the training workload received by each of the athletes.

This article is protected by copyright. All rights reserved.
Downloaded by: University of Cincinnati. Copyrighted material.
Accepted Manuscript

In view of the above, the idea that the ACWR is a good predictor of the risk of getting injured is ruled out since it has been proven to be an ineffective predictor on its own when detecting the occurrence of non-contact injuries in semiprofessional football players. However, the control of the training workload through the use of sRPE and obtaining the ACWR has been defined as a risk factor when football players get a non-contact injury [43]. Therefore, the control of said risk factor by technical staff may be an option so as to try to reduce the number of injuries suffered. Consequently, the control of such risk factor by the coaching staff may be an option to try to reduce the number of injuries suffered, especially in semiprofessional environments where the use of a free tool, such as the sRPE is a point in favour of its use since the economic possibilities of this type of club are lower than those of professional clubs and they cannot afford more expensive technologies such as GPS.

The main limitation of the study is that the number of injuries registered was low, especially if we compared with similar studies that analysed this topic. This makes that our conclusions should be taken with caution. In addition, the sample size was small, with only a team. However, most of studies in this topic also reported the ACWR analysis with only one team, in different TIER levels. Regarding this, the semiprofessional nature of the team allowed the development of other sports activities outside training and match hours, and the daily work of each player could be more or less physically demanding, which may or may not have influenced their workload and fatigue levels.

Conclusion

The aim of this study was to analyse the association and predictive capacity between the ACWR workload ratio and non-contact injuries in a semiprofessional football team. The results reject the idea that the value of the ACWR is correlated with the risk of non-contact injuries. Similarly, nothing has been found to support the ability to predict the occurrence of non-contact injuries using the ACWR workload ratio, perhaps because injuries are

This article is protected by copyright. All rights reserved.
Downloaded by: University of Cincinnati. Copyrighted material.
Accepted Manuscript

multifactorial (hormonal, neuromuscular, uncontrollable factors...) and cannot be reduced to a single factor.

In any case, the use of this tool in team planning and players management does not seem to be suitable as an injury prediction tool. Thus, other training load monitoring strategies are recommended to control and modulate the training process in team sports.

References

[1] Gusi N. Entrenamiento, preparaciones físicas y lesiones en el fútbol [Training, physical strength & conditioning and injuries in football]. Educ Física y Deport 2009; 21: 27–37

[2] López-Valenciano A, Ruiz-Pérez I, García-Gómez A, et al. Epidemiology of injuries in professional football: A systematic review and meta-analysis. Br J Sports Med 2020; 54: 711–718

[3] Football Research Group. UEFA Elite Club Injury Study: 2016/17 season report (2017). In Internet: https://www.uefa.com/MultimediaFiles/Download/uefaorg/Medical/02/49/97/62/2499762_DOWNLOAD.pdf ; (21 Feb 2023)

[4] Ekstrand J, Waldén M, Hägglund M. Hamstring injuries have increased by 4% annually in men's professional football, since 2001: A 13-year longitudinal analysis of the UEFA Elite Club injury study. Br J Sports Med 2016; 50: 731–737

[5] Hägglund M, Waldén M, Magnusson H, et al. Injuries affect team performance negatively in professional football: An 11-year follow-up of the UEFA Champions League injury study. Br J Sports Med 2013; 47: 738–742

[6] Hoffman DT, Dwyer DB, Bowe SJ, et al. Is injury associated with team performance in elite Australian football? 20 years of player injury and team performance data that include measures of individual player value. Br J Sports Med 2020; 54: 475–479

This article is protected by copyright. All rights reserved.

Accepted Manuscript

Downloaded by: University of Cincinnati. Copyrighted material.

[7] Jaspers A, Kuyvenhoven JP, Staes F, et al. Examination of the external and internal load indicators' association with overuse injuries in professional soccer players. J Sci Med Sport 2018; 21: 579–585. Available from: http://dx.doi.org/10.1016/j.jsams.2017.10.005

[8] Rossi A, Pappalardo L, Cintia P, et al. Effective injury forecasting in soccer with GPS training data and machine learning. PLoS One 2017; 13: 1–15

[9] Croisier JL, Ganteaume S, Binet J, et al. Strength imbalances and prevention of hamstring injury in professional soccer players: A prospective study. Am J Sports Med 2008; 36 :1469–1475

[10] Grooms DR, Palmer T, Onate JA, et al. Soccer-specific warm-up and lower extremity injury rates in collegiate male soccer players. J Athl Train 2013; 48: 782–789

[11] Nédélec M, Halson S, Delecroix B, et al. Sleep Hygiene and Recovery Strategies in Elite Soccer Players. Sport Med 2015; 45: 1547–1559

[12] Borg G. Psychophysical bases of perceived exertion. Med Sci Sports Exerc 1982; 14: 377–381. Available from: http://journal.um-surabaya.ac.id/index.php/JKM/article/view/2203

[13] Impellizzeri FM, Rampinini E, Coutts AJ, et al. Use of RPE-based training load in soccer. Med Sci Sports Exerc 2004; 36: 1042–1047

[14] Haddad M, Stylianides G, Djaoui L, et al. Session-RPE method for training load monitoring: Validity, ecological usefulness, and influencing factors. Front Neurosci 2017; 11: 1–14

[15] Casamichana D, Castellano J, Calleja-Gonzalez J, et al. Relationship between indicators of training load in soccer players. J Strength Cond Res 2013; 27: 369–374

[16] Gabbett TJ. The training-injury prevention paradox: Should athletes be training smarter and harder? Br J Sports Med 2016; 50: 273–280

This article is protected by copyright. All rights reserved.

Accepted Manuscript

Downloaded by: University of Cincinnati. Copyrighted material.

[17] Hulin BT, Gabbett TJ, Lawson DW, et al. The acute: Chronic workload ratio predicts injury: High chronic workload may decrease injury risk in elite rugby league players. Br J Sports Med 2016; 50: 231–236

[18] Malone S, Owen A, Newton M, Mendes B, et al. The acute:chonic workload ratio in relation to injury risk in professional soccer. J Sci Med Sport 2017; 20: 561–565. Available from: http://dx.doi.org/10.1016/j.jsams.2016.10.014

[19] Bowen L, Gross AS, Gimpel M, et al. Spikes in acute:chronic workload ratio (ACWR) associated with a 5-7 times greater injury rate in English Premier League football players: A comprehensive 3-year study. Br J Sports Med 2020; 54: 731–738

[20] Blanch P, Gabbett TJ. Has the athlete trained enough to return to play safely? The acute:chronic workload ratio permits clinicians to quantify a player's risk of subsequent injury. Br J Sports Med 2016; 50: 471–475

[21] Impellizzeri FM, Woodcock S, Coutts AJ, et al. What Role Do Chronic Workloads Play in the Acute to Chronic Workload Ratio? Time to Dismiss ACWR and Its Underlying Theory. Sport Med 2021; 51: 581–592. Available from: https://doi.org/10.1007/s40279-020-01378-6

[22] Fanchini M, Rampinini E, Riggio M, et al. Despite association, the acute:chronic work load ratio does not predict non-contact injury in elite footballers. Sci Med Footb 2018; 2: 108–114. Available from: https://doi.org/10.1080/24733938.2018.1429014

[23] Griffin A, Kenny IC, Comyns TM, et al. The Association Between the Acute:Chronic Workload Ratio and Injury and its Application in Team Sports: A Systematic Review. Sport Med 2020; 50: 561–580. Available from: https://doi.org/10.1007/s40279-019-01218-2

[24] Impellizzeri FM, Tenan MS, Kempton T, et al. Acute: Chronic workload ratio: Conceptual issues and fundamental pitfalls. Int J Sports Physiol Perform 2020; 15: 907–913

This article is protected by copyright. All rights reserved.

Accepted Manuscript

Downloaded by: University of Cincinnati. Copyrighted material.

[25] McKay AKA, Stellingwerff T, Smith ES, et al. Defining Training and Performance Caliber: A Participant Classification Framework. Int J Sports Physiol Perform 2022; 17: 317–631

[26] Foster C, Florhaug JA, Franklin J, et al. A new approach to monitoring exercise training. J Strength Cond Res 2001; 15: 109–115

[27] Fanchini M, Ferraresi I, Petruolo A, et al. Is a retrospective RPE appropriate in soccer? Response shift and recall bias. Sci Med Footb 2017; 1: 53–59. Available from: http://dx.doi.org/10.1080/02640414.2016.1231411

[28] Williams S, West S, Cross MJ, et al. Better way to determine the acute: Chronic workload ratio? Br J Sports Med 2017; 51: 209–210

[29] Murray NB, Gabbett TJ, Townshend AD, et al. Calculating acute: Chronic workload ratios using exponentially weighted moving averages provides a more sensitive indicator of injury likelihood than rolling averages. Br J Sports Med 2017; 51: 749–754

[30] Hägglund M, Waldén M, Bahr R, et al. Methods for epidemiological study of injuries to professional football players: Developing the UEFA model. Br J Sports Med 2005; 39: 340–346

[31] Doege J, Ayres JM, Mackay MJ, et al. Defining Return to Sport: A Systematic Review. Orthop J Sport Med 2021; 9: 1–8

[32] Kent P, Hancock MJ. Interpretation of dichotomous outcomes: sensitivity, specificity, likelihood ratios, and pre-test and post-test probability. J Physiother 2016; 62: 231–233

[33] van Beijsterveldt AM, Stubbe JH, Schmikli SL, et al. Differences in injury risk and characteristics between Dutch amateur and professional soccer players. J Sci Med Sport 2015; 18: 145–149. Available from: http://dx.doi.org/10.1016/j.jsams.2014.02.004

This article is protected by copyright. All rights reserved.

Accepted Manuscript

Downloaded by: University of Cincinnati. Copyrighted material.

[34] Hägglund M, Waldén M, Ekstrand J. Injury recurrence is lower at the highest professional football level than at national and amateur levels: Does sports medicine and sports physiotherapy deliver? Br J Sports Med 2016; 50: 751–758

[35] Manson J, Rahlf AL, Groll A, et al. The interval between matches significantly influences injury risk in field hockey. Int J Sports Med 2022; 43: 262-268.

[36] Pillay L, Burgess D, van Rensburg DCJ, et al. The congested international Match calendar in football : views of 1055 professional male players. BMC Sports Sci Med Rehabil [Internet] 2022; 14: 1-7. Available from: https://doi.org/10.1186/s13102-022-00597-w

[37] Carling C, Mccall A, Le Gall F, et al. The impact of short periods of match congestion on injury risk and patterns in an elite football club. J Sports Med 2015; 0: 1-6.

[38] Alentorn-Geli E, Myer GD, Silvers HJ, et al. Prevention of non-contact anterior cruciate ligament injuries in soccer players. Part 2: A review of prevention programs aimed to modify risk factors and to reduce injury rates. Knee Surgery, Knee Surg Sport Traumatol Arthrosc 2009; 17: 859–879

[39] van Dyk N, Farooq A, Bahr R, et al. Hamstring and Ankle Flexibility Deficits Are Weak Risk Factors for Hamstring Injury in Professional Soccer Players: A Prospective Cohort Study of 438 Players Including 78 Injuries. Am J Sports Med 2018; 46: 2203–2210

[40] Arnason A, Sigurdsson SB, Gudmundsson A, et al. Risk Factors for Injuries in Football. Am J Sports Med 2004; 32: 5–16

[41] Yu B, Garrett WE. Mechanisms of non-contact ACL injuries. Br J Sports Med 2007; 41 747–51

[42] Gregson W, Di Salvo V, Varley MC, et al. Harmful association of sprinting with muscle injury occurrence in professional soccer match-play: A two-season, league wide exploratory investigation from the Qatar Stars League. J Sci Med Sport 2020; 23: 134–138. Available from: https://doi.org/10.1016/j.jsams.2019.08.289

This article is protected by copyright. All rights reserved.

Accepted Manuscript

Downloaded by: University of Cincinnati. Copyrighted material.

392[43] McCall A, Dupont G, Ekstrand J. Internal workload and non-contact injury: A one-
393season study of five teams from the UEFA Elite Club Injury Study. Br J Sports Med 2018; 52:
3941517–1522
395

This article is protected by copyright. All rights reserved.
Accepted Manuscript
Downloaded by: University of Cincinnati. Copyrighted material.

LIST OF FIGURES

Figure 1. Non-contact injuries and non-injuries distribution regarding ACWR values

This article is protected by copyright. All rights reserved.

Downloaded by: University of Cincinnati. Copyrighted material.

Accepted Manuscript

LIST OF TABLES

Table 1. Descriptive means of participants of the study.

Table 2. Summary of the injuries that occurred during the time period studied.

Table 3. Spearman correlation coefficient between the ACWR variables and non-contact injury

Table 4. Summary of the true positives, false positives, true negatives and false negatives obtained, as well as the sensitivity and specificity of the test.

This article is protected by copyright. All rights reserved.

Accepted Manuscript

Downloaded by: University of Cincinnati. Copyrighted material.

Table 1. Descriptive means of participants of the study.

Height (cm)	178,48 ± 5,05
Weight (kg)	75,83 ± 9,86
Age (years)	25,59 ± 4,57
Experience* (years)	8,12± 4,95

* Experience means years of practice in senior category, and was calculated as: ("player ´s age" – 18).

This article is protected by copyright. All rights reserved.

Accepted Manuscript

Downloaded by: University of Cincinnati. Copyrighted material.

Table 2. Summary of the injuries that occurred during the time period studied.

Total number of injuries	15
Incidence of injuries (no. of injuries/1000h)	8.86
Injury severity (number of days out injured)	18,73 ± 19,81
Total number of non-contact injuries	10
Muscular	7 (70%)
Tendon	1 (10%)
Ligamentous	2 (20%)
Incidence of non-contact injuries (no. of injuries/1000h)	5,91
Non-contact injury severity (number of days out injured)	11,90 ± 10.38

This article is protected by copyright. All rights reserved.

Accepted Manuscript

Downloaded by: University of Cincinnati. Copyrighted material.

Table 3. Spearman correlation coefficient between the ACWR variables and non-contact injury.

			Injury
Spearman´s Rho	ACWR	Correlation coefficient	,069**
		N	2565

N = Number of times the value of the ACWR ratio was collected during the period studied.
ACWR: acute:chronic workload ratio
**p<0.05

This article is protected by copyright. All rights reserved.

Downloaded by: University of Cincinnati. Copyrighted material.

Accepted Manuscript

Table 4. Summary of the true positives, false positives, true negatives and false negatives obtained, as well as the sensitivity and specificity of the test.

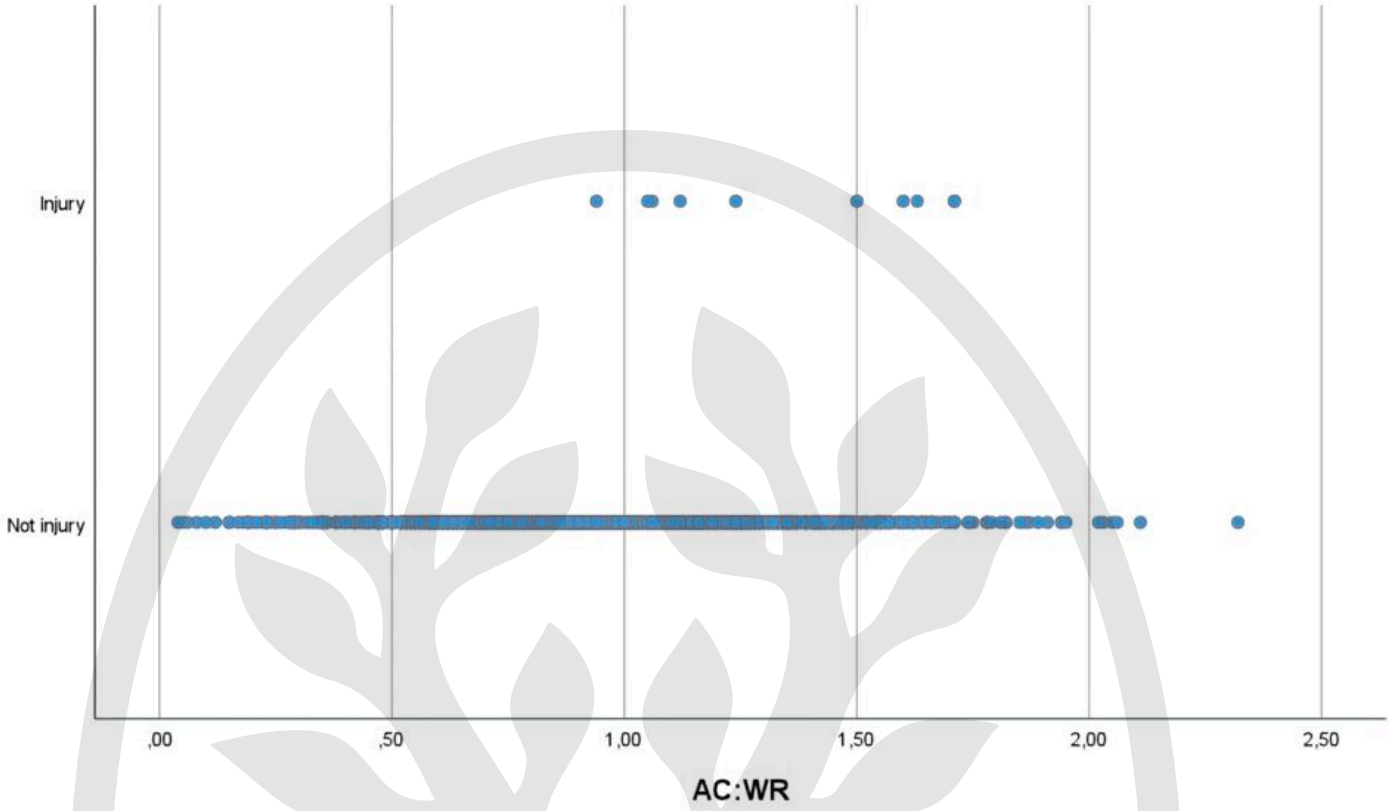
	True positives	False positives	True negatives	False negatives	Sensitivity	Specificity
ACWR	5	801	1754	5	50,00%	31,35%

ACWR: acute:chronic workload ratio

This article is protected by copyright. All rights reserved.

Downloaded by: University of Cincinnati. Copyrighted material.

Accepted Manuscript



This article is protected by copyright. All rights reserved.
Accepted Manuscript
Downloaded by: University of Cincinnati. Copyrighted material.